

On the longest linear rank-metric intersecting codes

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Abstract

Intersecting codes are a well-established class of linear codes in the Hamming metric, with deep connections to combinatorics, cryptography, and information theory. Recently, their analogue in the rank metric has been introduced, motivated by the growing relevance of rank-metric codes. Although the definition of rank-metric intersecting codes formally mirrors the Hamming-metric case, their structural properties and parameter constraints exhibit markedly different behavior. In a previous work, a geometric characterization of rank-metric intersecting codes was established, leading to the general parameter bounds $2k - 1 \leq n \leq 2m - 3$ for a rank-metric intersecting code of length n dimension k over $\mathbb{F}_{q^m}$. While the lower bound was shown to be tight, the tightness of the upper bound was left open. In this paper, we partially resolve this question by showing that the previously known upper bound can be attained only in the case $k = 3$ and $m \geq 6$, and that in all other cases the maximal value is strictly smaller, revealing a possible dependence on k .

Key Words : rank-metric code; intersecting code; scattered subspace

1 Introduction

Intersecting codes are a classical object in coding theory. In the Hamming metric, a linear code is said to be **intersecting** if the supports of any two

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nonzero codewords share at least one coordinate; [KS83, Mik84]. Since their introduction, intersecting codes have been studied extensively due to their rich combinatorial structure and their connections to several areas such as secret sharing schemes, oblivious transfer, separating systems, frameproof codes, and additive combinatorics; see e.g. [Mas93, BCS02, Bla03, Ran13, BSS25, PS11, Sco24]. In the binary case, intersecting codes coincide with minimal codes, a class that has attracted significant attention over the past decades.

In parallel, the theory of rank-metric codes has emerged as a central topic in modern coding theory, driven both by applications, most notably in network coding and post-quantum cryptography, and by deep connections with finite geometry; see [BHL⁺22, PZ20, She19a]. Rank-metric codes admit a natural geometric interpretation via q -systems, as shown in [Ran20] (see also [She19b, ABNR22]). This viewpoint has proven particularly fruitful in the study of extremal families such as MRD codes, minimal rank-metric codes, and related structures (see [ABNR22, ZZ21]).

Motivated by these developments, [BBMS25] has introduced and investigated the notion of **intersecting codes in the rank metric**. In this setting, a rank-metric code is said to be rank-metric intersecting if the rank supports of any two nonzero codewords intersect nontrivially. While this definition formally parallels the Hamming-metric case, the resulting class of codes exhibits substantially different behavior. In particular, rank-metric intersecting codes do not coincide with minimal rank-metric codes, although the two families intersect, and their existence and structure depend in a delicate way on the parameters of the code.

A key insight in the rank-metric setting is the geometric characterization of intersecting codes in terms of q -systems proved in [BBMS25]. More precisely, a rank-metric code is rank-metric intersecting if and only if its associated q -system is not generated by the sum of its intersections with two hyperplanes. This geometric viewpoint leads to nontrivial constraints on the parameters of rank-metric intersecting codes, provides new constructions beyond those arising from distance considerations alone, and reveals the presence of parameter ranges where the existence problem remains open. More precisely, the authors in [BBMS25] showed that for a rank-metric intersecting code in $\mathbb{F}_{q^m}^n$ of dimension k we have that

$$2k - 1 \leq n \leq 2m - 3.$$

The tightness of the lower bound was proved in the same paper, whereas the tightness of the upper bound was left as an open problem. In this paper, we show that the upper bound can only be reached if $k = 3$ and $m \geq 6$.

2 Preliminaries

2.1 Rank-metric codes and their geometry

Rank-metric codes were introduced by Delsarte [Del78] in 1978 as subsets of matrices and they have been intensively investigated in recent years because of their applications; we refer to [She19a, PZ20]. In this section we will be interested in rank-metric codes in $\mathbb{F}_{q^m}^n$. The **rank** (weight) $w(v)$ of a vector $v = (v_1, \dots, v_n) \in \mathbb{F}_{q^m}^n$ is defined as $w(v) = \dim_{\mathbb{F}_q}(\langle v_1, \dots, v_n \rangle_{\mathbb{F}_q})$.

A **(linear vector) rank-metric code** $\mathcal{C}$ is an $\mathbb{F}_{q^m}$ -subspace of $\mathbb{F}_{q^m}^n$ endowed with the rank distance, where such a distance is defined as $d(x, y) = w(x - y)$, where $x, y \in \mathbb{F}_{q^m}^n$. For details and applications we refer to [BHL⁺22, GR18].

Let $\mathcal{C} \subseteq \mathbb{F}_{q^m}^n$ be a linear rank-metric code. We will write that $\mathcal{C}$ is an $[n, k, d]_{q^m/q}$ code (or $[n, k]_{q^m/q}$ code) if $k = \dim_{\mathbb{F}_{q^m}}(\mathcal{C})$ and d is its minimum distance, that is $d = \min\{d(x, y) : x, y \in \mathcal{C}, x \neq y\}$. Moreover, for any $i \in \{1, \dots, n\}$, we define $A_i(\mathcal{C})$ the number of codewords in $\mathcal{C}$ having weight i .

By the classification of $\mathbb{F}_{q^m}$ -linear isometries of $\mathbb{F}_{q^m}^n$ (see [Ber03]), we say that two rank-metric codes $\mathcal{C}, \mathcal{C}' \subseteq \mathbb{F}_{q^m}^n$ are **(linearly) equivalent** if there exists a matrix $A \in \text{GL}(n, q)$ such that $\mathcal{C}' = \mathcal{C}A = \{vA : v \in \mathcal{C}\}$. The codes we will consider are **nondegenerate**, i.e. those for which the columns of any generator matrix of $\mathcal{C}$ are $\mathbb{F}_q$ -linearly independent.

The geometric counterpart of a rank-metric code is the so-called q -system. An $[n, k, d]_{q^m/q}$ **system** U is an $\mathbb{F}_q$ -subspace of $\mathbb{F}_{q^m}^k$ of dimension n , such that $\langle U \rangle_{\mathbb{F}_{q^m}} = \mathbb{F}_{q^m}^k$ and

$$d = n - \max \left\{ \dim_{\mathbb{F}_q}(U \cap H) \mid H \text{ is an } \mathbb{F}_{q^m}\text{-hyperplane of } \mathbb{F}_{q^m}^k \right\}.$$

Moreover, two $[n, k, d]_{q^m/q}$ systems U and U' are **equivalent** if there exists an $\mathbb{F}_{q^m}$ -isomorphism $\varphi \in \text{GL}(k, q^m)$ such that

$$\varphi(U) = U'.$$

Thanks to the following result, we can construct a system starting from a nondegenerate rank-metric code and conversely.

Theorem 2.1 ([Ran20, ABNR22]). Let $\mathcal{C}$ be a nondegenerate $[n, k, d]_{q^m/q}$ rank-metric code and let G be a generator matrix. Let $U \subseteq \mathbb{F}_{q^m}^k$ be the $\mathbb{F}_q$ -span of the columns of G . The rank weight of an element $xG \in \mathcal{C}$, with $x \in \mathbb{F}_{q^m}^k$ is

$$w(xG) = n - \dim_{\mathbb{F}_q}(U \cap x^\perp), \quad (1)$$

where $x^\perp = \{y \in \mathbb{F}_{q^m}^k : \langle x, y \rangle = 0\}$. In particular,

$$d = n - \max \left\{ \dim_{\mathbb{F}_q}(U \cap H) : H \text{ is an } \mathbb{F}_{q^m}\text{-hyperplane of } \mathbb{F}_{q^m}^k \right\}. \quad (2)$$

The above result allows us to give a one-to-one correspondence between equivalence classes of nondegenerate $[n, k, d]_{q^m/q}$ codes and equivalence classes of $[n, k, d]_{q^m/q}$ systems, see [Ran20]. The system U and the code $\mathcal{C}$ as in Theorem 2.1 are said to be **associated** and a system associated with a code $\mathcal{C}$ will usually be written as $U_{\mathcal{C}}$.

An essential concept is the rank support of a codeword. Let $\Gamma = (\gamma_1, \dots, \gamma_m)$ be an ordered $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^m}$. For any vector $x = (x_1, \dots, x_n) \in \mathbb{F}_{q^m}^n$ define the matrix $\Gamma(x) \in \mathbb{F}_q^{m \times n}$, where

$$x_j = \sum_{i=1}^m \Gamma(x)_{i,j} \gamma_i, \quad \text{for all } j \in \{1, \dots, n\},$$

that is $\Gamma(x)$ is the matrix expansion of the vector x with respect to the $\mathbb{F}_q$ -basis Γ of $\mathbb{F}_{q^m}$ and this clearly preserves its rank, i.e. $w(x) = \text{rk}(\Gamma(x))$. The **rank support** of x is defined as the column span of $\Gamma(x)$:

$$\text{supp}(x) = \text{rowsp}(\Gamma(x)) \subseteq \mathbb{F}_q^n.$$

As shown in [ABNR22, Proposition 2.1], the support of a vector is independent of Γ , allowing us to talk about the support of a word without reference to Γ . Let $\mathcal{C}$ be an $[n, k]_{q^m/q}$ code, we define the **support** of $\mathcal{C}$ as the vector sum of the supports of all the codewords of $\mathcal{C}$, i.e.

$$\text{supp}(\mathcal{C}) = \sum_{x \in \mathcal{C}} \text{supp}(x).$$

The support of the code $\mathcal{C}$ can be computed directly on $\mathcal{C}$, without using a fixed ordered basis.

2.2 Properties of $\mathbb{F}_q$ -subspaces in $\mathbb{F}_{q^m}^k$

It is well-known that $\mathbb{F}_{q^m}^k$ can be seen as an $\mathbb{F}_{q^m}$ -vector space of dimension k and as an $\mathbb{F}_q$ -vector space of dimension mk . The theory of linear sets is essentially based on the idea of studying $\mathbb{F}_q$ -subspaces of $\mathbb{F}_{q^m}^k$ and how they behave with respect to $\mathbb{F}_{q^m}$ -subspaces of $\mathbb{F}_{q^m}^k$. For instance, given an $\mathbb{F}_q$ -subspace U of $\mathbb{F}_{q^m}^k$ and W an $\mathbb{F}_{q^m}$ -subspace of $\mathbb{F}_{q^m}^k$, we define the **weight** of W in U as

$$\text{wt}_U(W) = \dim_{\mathbb{F}_q}(W \cap U).$$

A classical problem is the following: given the subspace U , what are the possible weights of the subspaces of a fixed dimension with respect to U ? We refer to [Pol10, LVdV15] for more details. In this section, we will rephrase some definitions and properties of linear sets in vector terms in order to fulfill the scope of the paper. We start with the definition of scattered subspaces,

originally introduced by Blokhuis and Lavrauw in [BL00]. An $\mathbb{F}_q$ -subspace U of $\mathbb{F}_{q^m}^k$ is said to be **scattered** if

$$\dim_{\mathbb{F}_q}(U \cap \langle v \rangle_{\mathbb{F}_{q^m}}) \leq 1,$$

for any $v \in \mathbb{F}_{q^m}^k$. The dimension of a scattered subspace is bounded as follows.

Theorem 2.2 ([BL00]). Let U be a scattered $\mathbb{F}_q$ -subspace in $\mathbb{F}_{q^m}^k$, then

$$\dim_{\mathbb{F}_q}(U) \leq \frac{mk}{2}.$$

When considering 2-dimensional $\mathbb{F}_{q^m}$ -subspaces $W = P_1 + P_2$ (where P_1 and P_2 are two distinct 1-dimensional $\mathbb{F}_{q^m}$ -subspaces), from a simple linear algebra argument we have that, for any $\mathbb{F}_q$ -subspace $U \subseteq W$,

$$\text{wt}_U(P_1 + P_2) \geq \text{wt}_U(P_1) + \text{wt}_U(P_2).$$

If the equality holds we say that U is with **complementary weights**. This definition was introduced in [NPSZ22] and extended in [AS24].

We conclude this section by describing the following duality operation, which preserves $\mathbb{F}_{q^m}$ -linearity when acting on the $\mathbb{F}_{q^m}$ -linear subspaces.

Let $\sigma: \mathbb{F}_{q^m}^k \times \mathbb{F}_{q^m}^k \rightarrow \mathbb{F}_{q^m}$ be a nondegenerate reflexive bilinear form over $\mathbb{F}_{q^m}$. Define $\sigma': \mathbb{F}_{q^m}^k \times \mathbb{F}_{q^m}^k \rightarrow \mathbb{F}_q$ by

$$\sigma' : (u, v) \mapsto \text{Tr}_{q^m/q}(\sigma(u, v)) = \sigma(u, v) + \sigma(u, v)^q + \dots + \sigma(u, v)^{q^{m-1}}.$$

If we regard $\mathbb{F}_{q^m}^k$ as an $\mathbb{F}_q$ -vector space, then σ' turns out to be a nondegenerate reflexive bilinear form on $\mathbb{F}_{q^m}^k$. Let $\perp$ and $\perp'$ be the orthogonal complement maps defined by σ and σ' on the lattices of $\mathbb{F}_{q^m}$ -linear and $\mathbb{F}_q$ -linear subspaces, respectively. The following properties hold (see [Pol10, Section 2] for more details).

Proposition 2.3. With the above notation,

- (i) $\dim_{\mathbb{F}_{q^m}}(W) + \dim_{\mathbb{F}_{q^m}}(W^\perp) = k$, for every $\mathbb{F}_{q^m}$ -subspace W of $\mathbb{F}_{q^m}^k$.
- (ii) $\dim_{\mathbb{F}_q}(U) + \dim_{\mathbb{F}_q}(U^{\perp'}) = mk$, for every $\mathbb{F}_q$ -subspace U of $\mathbb{F}_{q^m}^k$.
- (iii) $T_1 \subseteq T_2$ implies $T_1^{\perp'} \supseteq T_2^{\perp'}$, for every $\mathbb{F}_q$ -subspaces T_1, T_2 of $\mathbb{F}_{q^m}^k$.
- (iv) $W^\perp = (W^{\perp'})^{\perp'}$, for every $\mathbb{F}_{q^m}$ -subspace W of $\mathbb{F}_{q^m}^k$.
- (v) Let W and U be an $\mathbb{F}_{q^m}$ -subspace and an $\mathbb{F}_q$ -subspace of $\mathbb{F}_{q^m}^k$ of dimension s and t , respectively. Then

$$\dim_{\mathbb{F}_q}(U^{\perp'} \cap W^{\perp'}) - \dim_{\mathbb{F}_q}(U \cap W) = mk - t - sm,$$

i.e.

$$\text{wt}_{U^{\perp'}}(W^\perp) - \text{wt}_U(W) = mk - \dim_{\mathbb{F}_q}(U) - m \dim_{\mathbb{F}_{q^m}}(W). \quad (3)$$

3 Linear intersecting rank-metric codes

The notion of intersecting rank-metric code has been introduced in [BBMS25] as follows.

Definition 3.1. A code $\mathcal{C}$ is **rank-metric intersecting** if for any $c, c' \in \mathcal{C} \setminus \{0\}$ we have

$$\text{supp}(c) \cap \text{supp}(c') \neq \{0\}.$$

In order to describe the geometric counterpart of these codes, we give the following definition.

Definition 3.2. Let U be an $[n, k]_{q^m/q}$ system. We say that U is **2-spannable** if there exist two $\mathbb{F}_{q^m}$ -linear hyperplanes H_1 and H_2 in $\mathbb{F}_{q^m}^k$ such that

$$U = (H_1 \cap U) + (H_2 \cap U).$$

The authors in [BBMS25] proved that the geometric counterpart of an intersecting rank-metric code is a system that does not satisfy the 2-spannable property.

Theorem 3.3 (Theorem 3.4 in [BBMS25]). Let $\mathcal{C}$ be a nondegenerate $[n, k, d]_{q^m/q}$ code with a generator matrix G . The code $\mathcal{C}$ is rank-metric intersecting if and only if $U_{\mathcal{C}}$ is not 2-spannable.

Combining the geometric point of view and some results from [BSS25], the authors in [BBMS25] provided some bounds on the length of a rank-metric intersecting code.

Theorem 3.4 (Corollary 2.4 and Theorem 4.7 in [BBMS25]). Let $\mathcal{C}$ be a nondegenerate $[n, k, d]_{q^m/q}$ rank-metric intersecting code. Then

$$2k - 1 \leq n \leq 2m - 3.$$

Also, they showed constructions of rank-metric intersecting codes for a large sets of the parameters.

Theorem 3.5 (Theorem 4.4 in [BBMS25]). There exists a nondegenerate $[n, k, d]_{q^m/q}$ rank-metric intersecting code if

$$2k - 1 \leq n \leq 2m - 2k + 1.$$

Clearly, the natural question that the authors left open (see [BBMS25, Question 4.11]) regards the existence of these codes if the length is between $2m - 2k + 2$ and $2m - 3$, for $k \geq 3$.

4 Characterization of rank-metric intersecting codes of length $2m - 3$

We are going to give a geometric characterization of the rank-metric intersecting codes with maximum length $2m - 3$ in the case $k \geq 3$.

We start with the following characterization.

Proposition 4.1. Let U be an $\mathbb{F}_q$ -subspace of $\mathbb{F}_{q^m}^k$ of dimension n . The subspace U is not 2-spannable if and only if for any 2-dimensional $\mathbb{F}_{q^m}$ -subspace ℓ of $\mathbb{F}_{q^m}^k$ the linear set $\ell \cap U^{\perp'}$ is not complementary weight.

Proof. Clearly, U is not 2-spannable if and only if for any pairs of distinct $\mathbb{F}_{q^m}$ -linear hyperplanes H_1 and H_2 of $\mathbb{F}_{q^m}^k$ we have

$$(H_1 \cap U) + (H_2 \cap U) \subsetneq U.$$

Hence,

$$\text{wt}_U(H_1) + \text{wt}_U(H_2) - \text{wt}_U(H_1 \cap H_2) < n.$$

From Equation (3), the above inequality reads as follows

$$\text{wt}_{U^{\perp'}}(H_1^{\perp} + H_2^{\perp}) > \text{wt}_{U^{\perp'}}(H_1^{\perp}) + \text{wt}_{U^{\perp'}}(H_2^{\perp}),$$

i.e. the intersection of $U^{\perp'}$ with any 2-dimensional $\mathbb{F}_{q^m}$ -subspace of $\mathbb{F}_{q^m}^k$ is not complementary weight. $\square$

We are now going to focus on the case of length $2m - 3$.

Proposition 4.2. Let $\mathcal{C}$ be a nondegenerate $[2m - 3, k, d]_{q^m/q}$ rank-metric intersecting code with $k \geq 3$. Then $U_{\mathcal{C}}^{\perp'}$ is a scattered $\mathbb{F}_q$ -subspace.

Proof. Observe that $U_{\mathcal{C}}^{\perp'}$ is an $\mathbb{F}_q$ -subspace of $\mathbb{F}_{q^m}^k$ with dimension $km - 2m + 3$. By contradiction suppose that $U_{\mathcal{C}}^{\perp'}$ is not scattered. Then there exists $P = \langle v \rangle_{\mathbb{F}_{q^m}}$ with $v \in \mathbb{F}_{q^m}^k \setminus \{0\}$ such that $w_{U_{\mathcal{C}}^{\perp'}}(P) = h \geq 2$. Every 2-dimensional $\mathbb{F}_{q^m}$ -subspace of $\mathbb{F}_{q^m}^k$ through P has weight at least $h + 2$, otherwise the intersection between such a 2-dimensional $\mathbb{F}_{q^m}$ -subspace and $U_{\mathcal{C}}^{\perp'}$ is complementary weight, a contradiction to Proposition 4.1. This means that the number of nonzero vectors in $U_{\mathcal{C}}^{\perp'}$ is at least the number of vectors in

$$\bigcup_{W \in \mathcal{L}} (W \cap U_{\mathcal{C}}^{\perp'}) \setminus P \cup (P \cap U_{\mathcal{C}}^{\perp'}),$$

where $\mathcal{L}$ is the set of all the 2-dimensional $\mathbb{F}_{q^m}$ -subspaces of $\mathbb{F}_{q^m}^k$ through P . Therefore, we get

$$q^{(k-2)m+3} - 1 \geq \frac{q^{m(k-1)} - 1}{q^m - 1} (q^{h+2} - q^h) + q^h - 1,$$

from which we derive

$$q^{m(k-2)+3-h} \geq 1 + \frac{q^{m(k-1)} - 1}{q^m - 1}(q^2 - 1),$$

and so

$$q^{m(k-1)} - q^{m(k-1)+1-h} + q^{m(k-2)+1-h} - q^{m(k-1)-2} + q^{m-2} - 1 \leq 0,$$

a contradiction. $\square$

Theorem 4.3. Let $\mathcal{C}$ be a nondegenerate $[2m - 3, k, d]_{q^m/q}$ rank-metric intersecting code with $k \geq 3$. Then $k = 3$ and $m \geq 6$.

Proof. Let $U_{\mathcal{C}}$ be a system associated with $\mathcal{C}$. By Proposition 4.2, we know that $U_{\mathcal{C}}^{\perp'}$ is a scattered $\mathbb{F}_q$ -subspace of dimension $mk - 2m + 3$ in $\mathbb{F}_{q^m}^k$. Therefore, by Theorem 2.2 we have that

$$mk - 2m + 3 \leq \frac{mk}{2},$$

from which we derive that

$$k \leq 4 - \frac{6}{m},$$

and so the assertion. $\square$

Therefore, we have the following characterization for rank-metric intersecting code of length $2m - 3$.

Corollary 4.4. Let $\mathcal{C}$ be a nondegenerate $[2m - 3, k, d]_{q^m/q}$ rank-metric code with $k \geq 3$. The code $\mathcal{C}$ is rank-metric intersecting if and only if the following three conditions hold

- $k = 3$;
- $m \geq 6$;
- $U_{\mathcal{C}}^{\perp'}$ is a scattered subspace.

Proof. The necessary condition follows from Proposition 4.2 and Theorem 4.3. For the converse, observe that if $U_{\mathcal{C}}^{\perp'}$ is a scattered subspace of dimension $m + 3$ in $\mathbb{F}_{q^m}^3$, then every two-dimensional subspace $\ell \subseteq \mathbb{F}_{q^m}^3$ intersects $U_{\mathcal{C}}^{\perp'}$ in a scattered subspace of dimension at least 3. In particular, the intersection $\ell \cap U_{\mathcal{C}}^{\perp'}$ is not complementary weight. By Proposition 4.2, it follows that $U_{\mathcal{C}}$ is not 2-spannable. The claim now follows from Theorem 3.3. $\square$

5 Some existence results

In this section we deal with existence results of nondegenerate $[2m-3, k, d]_{q^m/q}$ rank-metric intersecting code with $k \geq 3$. As it has been shown in Theorem 4.3, we need that $k = 3$ and $m \geq 6$. Therefore, by Corollary 4.4, the existence problem of a $[2m-3, 3, d]_{q^m/q}$ rank-metric intersecting code with $m \geq 6$ is now translated into the existence of a scattered $\mathbb{F}_q$ -subspace of $\mathbb{F}_{q^m}^3$ of dimension $m+3$. Hence, we are going to recall some existence results for scattered spaces, and we will derive some existence results on the related rank-metric intersecting codes.

If m is even, we know that scattered $\mathbb{F}_q$ -subspaces of dimension $3m/2$ in $\mathbb{F}_{q^m}^3$ always exist as a consequence of the results in [BGMP18, CMPZ17].

Theorem 5.1 ([BGMP18, CMPZ17]). Suppose that m is even. Then there exists a scattered $\mathbb{F}_q$ -subspace of dimension $3m/2$ in $\mathbb{F}_{q^m}^3$.

If m is odd, very few is known.

As a consequence of Corollary 4.4 and Theorem 5.1, we obtain the following existence result.

Corollary 5.2. Suppose that $m \geq 6$ is even. There exists a nondegenerate $[2m-3, 3, d]_{q^m/q}$ rank-metric intersecting code.

This shows that the upper bound in Theorem 3.4 is achieved for $k = 3$ and $m \geq 6$.

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References

- [ABNR22] Gianira N Alfarano, Martino Borello, Alessandro Neri, and Alberto Ravagnani. Linear cutting blocking sets and minimal codes in the rank metric. *Journal of Combinatorial Theory, Series A*, 192:105658, 2022.
- [AS24] Sam Adriaensen and Paolo Santonastaso. On the minimum size of linear sets. *The Electronic Journal of Combinatorics*, pages P4–52, 2024.

- [BBMS25] Daniele Bartoli, Martino Borello, Giuseppe Marino, and Martin Scotti. Linear rank-metric intersecting codes. *arXiv preprint arXiv:2507.00569*, 2025.
- [BCS02] Gilles Brassard, Claude Crépeau, and Miklos Santha. Oblivious transfers and intersecting codes. *IEEE Transactions on Information Theory*, 42(6):1769–1780, 2002.
- [Ber03] Thierry P Berger. Isometries for rank distance and permutation group of Gabidulin codes. *IEEE Transactions on Information Theory*, 49(11):3016–3019, 2003.
- [BGMP18] Daniele Bartoli, Massimo Giulietti, Giuseppe Marino, and Olga Polverino. Maximum scattered linear sets and complete caps in Galois spaces. *Combinatorica*, 38(2):255–278, 2018.
- [BHL⁺22] Hannes Bartz, Lukas Holzbaur, Hedongliang Liu, Sven Puchinger, Julian Renner, Antonia Wachter-Zeh, et al. Rank-metric codes and their applications. *Foundations and Trends® in Communications and Information Theory*, 19(3):390–546, 2022.
- [BL00] Aart Blokhuis and Michel Lavrauw. Scattered spaces with respect to a spread in $PG(n, q)$. *Geometriae Dedicata*, 81(1):231–243, 2000.
- [Bla03] Simon R Blackburn. Frameproof codes. *SIAM Journal on Discrete Mathematics*, 16(3):499–510, 2003.
- [BSS25] Martino Borello, Wolfgang Schmid, and Martin Scotti. The geometry of intersecting codes and applications to additive combinatorics and factorization theory. *Journal of Combinatorial Theory, Series A*, 214:106023, 2025.
- [CMPZ17] Bence Csajbók, Giuseppe Marino, Olga Polverino, and Ferdinando Zullo. Maximum scattered linear sets and MRD-codes. *Journal of Algebraic Combinatorics*, 46(3):517–531, 2017.
- [Del78] Philippe Delsarte. Bilinear forms over a finite field, with applications to coding theory. *Journal of Combinatorial Theory, Series A*, 25(3):226–241, 1978.
- [GR18] Elisa Gorla and Alberto Ravagnani. Codes endowed with the rank metric. *Greferath M., Pavčević M., Silberstein N., Vázquez-Castro M. (eds) Network Coding and Subspace Designs. Signals and Communication Technology.*, 2018.

- [KS83] Gyula Katona and Jaya Srivastava. Minimal 2-coverings of a finite affine space based on $\text{GF}(2)$. *Journal of statistical planning and inference*, 8(3):375–388, 1983.
- [LVdV15] Michel Lavrauw and Geertrui Van de Voorde. Field reduction and linear sets in finite geometry. *Topics in finite fields*, 632:271–293, 2015.
- [Mas93] James L Massey. Minimal codewords and secret sharing. In *Proceedings of the 6th joint Swedish-Russian international workshop on information theory*, pages 276–279, 1993.
- [Mik84] Dezso Miklós. Linear binary codes with intersection properties. *Discrete applied mathematics*, 9(2):187–196, 1984.
- [NPSZ22] Vito Napolitano, Olga Polverino, Paolo Santonastaso, and Ferdinando Zullo. Linear sets on the projective line with complementary weights. *Discrete Mathematics*, 345(7):112890, 2022.
- [Pol10] Olga Polverino. Linear sets in finite projective spaces. *Discrete Mathematics*, 310(22):3096–3107, 2010.
- [PS11] Alain Plagne and Wolfgang A Schmid. An application of coding theory to estimating Davenport constants. *Designs, Codes and Cryptography*, 61(1):105–118, 2011.
- [PZ20] Olga Polverino and Ferdinando Zullo. Connections between scattered linear sets and MRD-codes. *Bulletin of the Institute of Combinatorics and its Applications*, 89:46–74, 2020.
- [Ran13] Hugues Randriambololona. $(2, 1)$ -separating systems beyond the probabilistic bound. *Israel Journal of Mathematics*, 195(1):171–186, 2013.
- [Ran20] Tovohery Hajatiana Randrianarisoa. A geometric approach to rank metric codes and a classification of constant weight codes. *Designs, Codes and Cryptography*, 88:1331–1348, 2020.
- [Sco24] Martin Scotti. Recent advances on minimal codes. *arXiv preprint arXiv:2411.11882*, 2024.
- [She19a] John Sheekey. MRD codes: constructions and connections. *Combinatorics and Finite Fields: Difference Sets, Polynomials, Pseudorandomness and Applications*, 23, 2019.
- [She19b] John Sheekey. (Scattered) Linear Sets are to Rank-Metric Codes as Arcs are to Hamming-Metric Codes. In Marcus Greferath, Camilla Hollanti, and Joachim Rosenthal, editors, *Oberwolfach Report No. 13/2019*, 2019.

- [ZZ21] Giovanni Zini and Ferdinando Zullo. Scattered subspaces and related codes. *Designs, Codes and Cryptography*, 89(8):1853–1873, 2021.